

KENDRIYA VIDYALAYA SANGATHAN ERNAKULAM REGION**PRE-BOARD EXAMINATION-2024 -25****Class: XII PHYSICS****ANSWER KEY**

Q.NO	VALUE POINTS	MARKS
1	C	1
2	C	1
3	C	1
4	C	1
5	A	1
6	C	1
7	A	1
8	A	1
9	C	1
10	B	1
11	C	1
12	B	1
13	B	1
14	A	1
15	D	1
16	C	1
17	a) UV radiations b) Radio waves c) Infra red radiations Radio waves > Infra red radiations > UV radiations	$\frac{1}{2} + \frac{1}{2} +$ $\frac{1}{2} + \frac{1}{2}$
18	$R = \rho \frac{l}{A} = 4 \rho \frac{l}{\pi D^2}$	$\frac{1}{2}$

	$R_1/R_2 = \frac{1}{2} \times (3/2)^2 = 9/8$ Since current remains the same, $V_1/V_2 = R_1/R_2 = 9/8$	1 $\frac{1}{2}$
19	$B_1 = \mu_0 I_1 / 2R$ $\Phi_2 = B_1 A_2 = \mu_0 I_1 / 2R \times \pi r^2 = M_{21} I_1$ $M_{21} = \mu_0 \pi r^2 / 2R$ OR $P_{in} = E_p I_p$ or $I_p = P_{in} / E_p = 1100/220 = 5A$ $E_s = kE_p = 100 \times 220 = 22000V$	$\frac{1}{2}$ 1 $\frac{1}{2}$ 1 1
20	$E_2 - E_1 = K (1/1^2 - 1/2^2) = K \frac{3}{4}$ $E_\infty - E_2 = K (1/2^2 - 1/\infty) = k \frac{1}{4}$ Ratio = 3:1	1 1
21	Circuit diagram + explanation	1 + 1
22	For series combination $C = C_x C_y / (C_x + C_y)$ $4 \mu F = 4C/5$, $C = 5 \mu F$, $C_x = 5 \mu F$, $C_y = 20 \mu F$ Total charge $Q = CV = 4 \mu F \times 15V = 60 \mu C$ $V_x = Q / C_x = 60/5 = 12 V$ $V_y = Q / C_y = 60/20 = 3 V$ $E_x / E_y = (\frac{1}{2} Q_x^2 / C_x) / (\frac{1}{2} Q_y^2 / C_y) = 20/5 = 4:1$	$\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
23	Kirchhoff's laws $I_3 = I_1 + I_2$	$\frac{1}{2}$ $\frac{1}{2}$

	For upper loop, $3 I_2 - 4 I_1 = 1$	$\frac{1}{2}$
	For 2 nd loop, $3 I_2 + 2 I_3 = 3$	$\frac{1}{2}$
	Solving, $I_1 = 2/13 \text{ A}$, $I_2 = 5/13 \text{ A}$, $I_3 = 7/13 \text{ A}$	1
	OR	
	For AFEBA, $(1+1) I = 3$, $I = 1.5 \text{ A}$	1
	For BEDCB, $1.5 R + 4 \times 0 = 3$, $R = 2\Omega$	1
	$V_{AD} = (1+1) \times 1.5 = 3\text{V}$	1
24	Biot-Savart law	$\frac{1}{2}$
	Derivation of B	2
	Mag. field at the centre	$\frac{1}{2}$
25	Induced emf = $NBA\omega \sin \omega t$	$\frac{1}{2}$
	Induced emf is same in both loops (B, A and ω are same for both loops)	1
	The resistivity of copper is less than aluminium	1
	therefore current induced is larger in copper loop	$\frac{1}{2}$
26	For constructive interference, the distance between the two fringes is given by relation: $x = n\lambda D/d$ where, n = Order of fringes	1
	$\lambda = xd/nD = 1.2 \times 10^{-2} \times 0.28 \times 10^{-3} / 4 \times 1.4 = 6 \times 10^{-7} \text{ m}$	2
27	The threshold frequency for metal X is less and hence threshold wavelength of metal X will be larger.	1
	As threshold frequency for X is less than that of Y and work function $\phi_0 = h\nu_0$, hence work function for X is less than that of Y. Lesser the work function, more will be its K.E.	1
	(K.E.) $\text{max} = h\nu - \phi_0$, K.E. of X will be more.	
	(iii) If the distance between the light source and metal X is halved, frequency of light remains unchanged. Therefore, the K.E. of ejected electron remains unchanged.	1

28	Graph + any 2 points One example each for fission & fusion	1 +1 ½+ ½
29	1. B 2. D 3. B 4. B or C	1 1 1 1
30	1 A or B 2 C 3. C 4.B	1 1 1 1
31	a) Statement & expression (b) derivation of electric field due to uniformly charged infinite plane sheet. (c) $E = \frac{\sigma}{\epsilon_0} = 3 \text{ N/C}$ OR a) Derivation of electric field at axial line of electric dipole b) Resultant dipole moment is given as $R = (P^2 + P^2 + 2PP\cos 120^\circ)^{1/2} = P$	½+ ½ 2 1+ 1 3 2
32	Derivation & definition of Ampere $F = \mu_0 I_1 I_2 l / 2\pi r$ Substitution & calculation $I = 0.5 \text{ A}$ OR Principle, construction and working $S = I_g G/I - I_g = 10^{-3} \times 10 / (100-1) 10^{-3} = 1\Omega (\text{approx.})$	2 + 1 ½ 1½ 3 2

33	Ray diagram Expression for magnifying power $1/v_0 - 1/u_0 = 1/f_0$ $1/v_0 = 1/1.25 - 1/1.5 = 2/15 \quad V_0 = 7.5 \text{ cm}$ $m = v_0/u_0 (1 + d/f_e)$ $m = 7.5/1.5(1 + 25/5) = 30$ OR Diagram + Derivation $f_a = 100/10 = 10 \text{ cm}$ $1/f_a = (\mu_g - 1)(1/R_1 - 1/R_2)$ $1/f_1 = (\mu_g/\mu_l - 1)(1/R_1 - 1/R_2)$ Solving, $\mu = 1.67$	2 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2} + 2\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
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